

TAIKOCIN 30% INJECTION

Description and Composition

TAIKOCIN 30% INJECTION is a light yellow to yellow clear solution. Each ml contains 300mg of Florfenicol as its active ingredient.

Pharmacodynamics

Florfenicol is a synthetic broad-spectrum antibiotic effective against most Gram positive and Gram negative bacteria isolated from domestic animals. Florfenicol acts by inhibiting bacteria proteins synthesis at the ribosomal level, thus is bacteriostatic. However, *in vitro* tests have shown that florfenicol has a bactericidal activity against the most commonly isolated bacterial pathogens involved in respiratory diseases:

- *Histophilus somni*, *Mannheimia haemolytica* and *Pasteurella multocida* isolated from cattle
- *Actinobacillus pleuropneumonia* and *Pasteurella multocida* isolated from pigs.

Acquired resistance to florfenicol is mediated by efflux pump resistance associated with a *floR* gene. Such resistance has not yet been identified in the target pathogens except for *Pasteurella multocida*. Cross resistance with chloramphenicol can occur. Resistance to florfenicol and other antimicrobials has been identified in the food-born pathogen *Salmonella typhimurium* and co-resistance with the third-generation cephalosporins has been observed in respiratory and digestive *Escherichia coli*. For *Mannheimia haemolytica*, *Pasteurella multocida* and *Histophilus somni* the following breakpoints have been determined for florfenicol in bovine respiratory disease; susceptible: ≤ 2 µg/ml, intermediate: 4 µg/ml, resistant: ≥ 8 µg/ml.

In bovine, 99% of *P. multocida* isolates (n=156) and 98% of *M. haemolytica* isolates (n=109) were susceptible to florfenicol (strains isolated in France in 2012). In swine, 99% of *P. multocida* isolates (n=150) were susceptible to florfenicol (strains isolated in France in 2012).

Pharmacokinetics

In cattle:

Intramuscular administration of the formulation at the recommended dose of 20 mg/kg maintains efficacious blood levels for 48 hours. Maximum mean serum concentration (C_{max}) of 3.8 µg/ml occurred 5.7 hours (T_{max}) after dosing. The mean serum concentration 24 hours after dosing was 1.95 µg/ml. The mean elimination half life was 15.3 hours.

In pigs:

After intramuscular administration of florfenicol, maximum serum concentration of 4.7 µg/ml is reached after 1.8 hours and the concentrations deplete with a terminal mean half-life of 14.8 hours. Serum concentrations drop below 1 µg/ml, the MIC₉₀ for the target porcine pathogens, 12-24 hours following IM administration. Florfenicol concentrations achieved in lung tissue reflect plasma concentration, with a lung: plasma concentration ratio of approximately 1. After administration to pigs by the intramuscular route, florfenicol is rapidly excreted, primarily in urine. The florfenicol is extensively metabolised.

Indication

Cattle: Treatment of respiratory tract infections due to *Mannheimia haemolytica*, *Pasteurella multocida* and *Histophilus somni* susceptible to florfenicol.

Pigs: Treatment of acute outbreaks of swine respiratory disease caused by strains of *Actinobacillus pleuropneumoniae* and *Pasteurella multocida* susceptible to florfenicol.

Recommended Dose

By intramuscular injection into the neck.

Cattle: 20mg florfenicol per kg body weight, i.e. 1ml of solution per 15kg body weight, twice 48 hours apart.

Pigs: 15mg florfenicol per kg body weight, i.e. 1ml of solution per 20kg body weight, twice 48 hours apart.

The dose volume given at any one injection site should not exceed 10ml in cattle and 3ml in pigs. To ensure a correct dosage body weight should be determined as accurately as possible to avoid underdosing.

It is recommended to treat animals in the early stages of disease and to evaluate the response to treatment within 48 hours after the second injection. If clinical signs of respiratory disease persist 48 hours after the last injection, treatment should be changed using another formulation or another antibiotic and continued until clinical signs have resolved.

Route of Administration

To be given by intramuscular injection in cattle and pigs.

Contraindications

Do not use in adult bulls or boars intended for breeding purposes.

Do not administer in cases of hypersensitivity to the active ingredient or any of the excipients.

Warning and Precautions

Special warnings for each target species:

None.

Special precautions for use

Special precautions for use in animals:

Wipe the stopper before removing each dose. Use a dry, sterile needle and syringe. Do not use in piglets of less than 2kg.

Under field conditions, approximately 30% of treated pigs presented with pyrexia (40°C) associated with either moderate depression or moderate dyspnoea a week or more after administration of the second dose.

Use of product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria. Official national and regional antimicrobial policies should be taken into account when the product is used.

Use of the product deviating from the instructions may increase the prevalence of bacteria resistant to the florfenicol and may decrease the effectiveness of treatment with other antimicrobials, due to the potential for cross-resistance. Particular attention should be paid to improving farming practices to avoid any stress condition (improving management practices and by cleaning and disinfection).

Special precautions to be taken by the person administering the veterinary medicinal product to animals:

Care should be taken when handling the product to avoid accidental self-injection. In case of accidental self-injection, seek medical advice, and show the package leaflet or the label to the physician.

People with known hypersensitivity to the components of the formulation should avoid contact with the product.

Wash hands after handling the product.

Interaction with Other Medicaments

None known.

Pregnancy and Lactation

Studies in laboratory animals have not revealed any evidence of embryo- or foetotoxic potential for florfenicol. However, the safety of florfenicol on bovine and porcine reproductive performance and pregnancy has not been assessed. Use only according to the benefit/risk assessment by the responsible veterinarian.

Side Effects

Cattle:

A decrease in food consumption and transient softening of the faeces may occur during the treatment period. Treated animals recover quickly and completely upon termination of treatment.

Administration of the product by the intramuscular route may cause inflammatory lesions at the injection site which may persist for up to 28 days.

Pigs:

Commonly observed adverse effects are transient diarrhoea and/or peri-anal and rectal erythema/oedema which may affect 50% of the animals. These effects may be observed for up to one week. Administration of the product by the intramuscular route may cause inflammatory lesions at injection site which disappear within 28 days.

Symptoms and Treatment of Overdose

In swine after intramuscular administration of 3 times the recommended dose or more, a reduction in feeding, hydration and weight gain has been observed. After administration of 5 times the recommended dose or more, vomiting has also been noted.

Storage Condition

Store below 30°C.

Shelf life

3 years.

Following withdrawal of first dose, use the product within 24 hours. Any unused material should be discarded.

Withdrawal Periods

Cattle:

Meat and offal: 37 days

Milk: Not permitted for use in lactating animals producing milk for human consumption.

Pigs:

Meat and offal: 18 days

Packing

20ml, 50ml, 100ml, 250ml.

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	Manufacturer & Product Registration Holder
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